Inference in OLS

These notes accompany Lecture 3. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Learning outcomes – after this lecture you should be able to:

1. State assumption MLR.6 and explain what the CLM assumptions add to Gauss-Markov.
2. Derive and use the t-statistic to test a hypothesis about a single parameter, against one- and two-sided alternatives.
3. Interpret a p-value correctly, and explain why it is more informative than a fixed significance level.
4. Distinguish statistical significance from economic significance.
5. Construct and interpret a confidence interval, and use it to conduct a two-sided test.
6. Test a hypothesis about a linear combination of two parameters.
7. Use the F-test for multiple restrictions, and explain why variables can be jointly significant yet individually insignificant.

Contents

1. Sampling distributions of the OLS estimators
2. Testing hypotheses about a single parameter: the t-test
3. Confidence intervals
4. Testing a single linear combination of the parameters
5. Testing multiple linear restrictions: the F-test
6. Reporting regression results
7. Summary

Appendix. Asymptotic normality and large-sample inference

1. Sampling distributions of the OLS estimators

1.1 Why unbiasedness is not enough

So far we have assembled a set of assumptions under which OLS is unbiased and has the smallest variance among linear unbiased estimators. That tells us where the estimator is centred and how tightly it is spread – but not what shape its distribution has.

To test hypotheses about the parameters and to construct confidence intervals, we need the full sampling distribution of the estimators, not just their mean and variance.

1.2 Assumption MLR.6: normality

The population error *u* is independent of the explanatory variables x_1, x_2, …, x_k and is normally distributed with zero mean and variance σ^2:

$$u \sim Normal(0, \sigma^2)$$

MLR.6 is much stronger than any previous assumption. Note carefully what it subsumes: if you assume MLR.6, you are necessarily assuming MLR.4 (zero conditional mean) and MLR.5 (homoskedasticity) as well, because independence delivers both.

In cross-sectional settings, assumptions MLR.1 through MLR.6 together are called the **classical linear model (CLM)** assumptions.

Assumption set	What it buys you
MLR.1 – MLR.4	OLS is unbiased
MLR.1 – MLR.5 (Gauss-Markov)	OLS is BLUE – smallest variance among linear unbiased estimators
MLR.1 – MLR.6 (CLM)	OLS is normally distributed; exact t and F inference is valid; OLS is the minimum-variance unbiased estimator among all unbiased estimators, not just linear ones

Summarised in population terms: conditional on x, y has a normal distribution with a mean linear in the regressors and a constant variance.

Is normality plausible?

Whether normality of u can be assumed is an empirical matter, and often the answer is no. There is no theorem saying that wage, conditional on education, experience and tenure, is normally distributed. Simple reasoning suggests the opposite: wage can never be negative, so strictly speaking it cannot be normal. Minimum wage laws mean some fraction of the population earns exactly the minimum, which also violates normality.

Often a transformation – especially taking logs – yields a distribution closer to normal. And in large samples the assumption can be dropped altogether; see the Appendix.

1.3 Normal sampling distributions

Under MLR.1 through MLR.6, conditional on the sample values of the independent variables:

$$\hat{\beta}_j \sim \text{Normal}[\beta_j, \text{Var}(\hat{\beta}_j)]$$

Standardising gives a standard normal variable:

$$\frac{\hat{\beta}_j - \beta_j}{sd(\hat{\beta}_j)} \sim Normal(0,1)$$

But sd($\hat{\beta}_j$) depends on the unknown σ. Replacing it with the estimated standard error costs us the normal distribution and gives the t-distribution instead:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1} = t_{df}$$

where *k* is the number of slope parameters and $n - k - 1$ is the **degrees of freedom** (df).

> **Why the t-distribution, not the normal?**
>
> The only difference between the two lines above is that the unknown σ has been replaced by an estimate. That extra estimation adds uncertainty, which the t-distribution accounts for through its fatter tails.
>
> As df grows the t-distribution converges to the standard normal – which is why the rule of thumb in Section 2.5 works for large samples.

2. Testing hypotheses about a single parameter: the t-test

> **Read this before anything else in the section**
>
> We are testing hypotheses about population parameters – not about the estimates from a particular sample.
>
> The β_j are unknown features of the population and we will never know them with certainty. We can hypothesise about their values and use statistical inference to assess those hypotheses.

2.1 The null hypothesis

In many applications the hypothesis of interest is:

$$H_0: \beta_j = 0$$

This says that once all the other regressors have been accounted for, x_j has no effect on the expected value of y.

Take the wage equation:

$$log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

Here H_0: $\beta_2 = 0$ means that once education and tenure are accounted for, years in the workforce have no effect on hourly wage – that is, a person's work history prior to their current employment does not affect their wage. That is an economically interesting claim. If instead $\beta_2 > 0$, prior work experience contributes to productivity and hence to wage.

2.2 One-sided alternatives

Consider the one-sided alternative $H_1: \beta_j > 0$ (or $H_1: \beta_j < 0$). The rejection rule has four steps:

1. Decide on the significance level α – the probability of rejecting H_0 when it is in fact true. The 5% level ($\alpha = 0.05$) is the most common choice.
2. Compute the t-statistic, $t = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j)$, which under $H_0: \beta_j = 0$ reduces to $t = \hat{\beta}_j/se(\hat{\beta}_j)$. It measures how many estimated standard errors the estimate lies from the hypothesised value.
3. Find the critical value c for $n - k - 1$ degrees of freedom from the t-table (or from software).
4. For $H_1: \beta_j > 0$, reject H_0 if $t > c$. For $H_1: \beta_j < 0$, reject H_0 if $t < -c$.

This is a one-tailed test: the whole rejection region sits in one tail. Notice that the sign of t matters here – a large negative t is not evidence for $H_1: \beta_j > 0$.

Wage example

Using the 526 observations:

$$\textbf{log(wage) = 0.284 + 0.092 educ + 0.004 exper + 0.022 tenure}$$

Test whether the return to experience, controlling for education and tenure, is zero against the alternative that it is positive: H_0: β_exper = 0 versus H_1: β_exper > 0.

Quantity	Value
Standard error (reported by EViews)	0.001723
t-statistic on exper (reported by EViews)	2.39 (0.004121 / 0.001723)
Degrees of freedom ($n - k - 1 = 526 - 3 - 1$)	522
Critical value at 5%, one-sided	1.648
Decision	Reject H_0 – β_exper is statistically greater than zero at the 5% level

The t-statistic for our sample parameter is given by EViews and equals 2.39. The critical value for 522 degrees of freedom is 1.648. The t-stat is higher than its critical value, so at the 5% significant level we reject the null and can conclude, that β_{exper} is statistically greater than zero. We have persuasively shown that the partial effect of experience on wage is positive in the population.

In EViews the critical value is obtained with @qtdist(1 − α, df) – here @qtdist(0.95, 522).

2.3 Two-sided alternatives

In most applications we test against a two-sided alternative, $H_1: \beta_j \neq 0$. This says x_j has a ceteris paribus effect on y without specifying its direction. Because the direction is unspecified, we care about the absolute value of t:

$$| t | > c$$

For a two-tailed test, c is chosen so that each tail of the t-distribution holds $\alpha/2$. At the 5% level with df = 25, c = 2.060.

- **If H_0 is rejected** at the 5% level we say β_j is "statistically significant", or "statistically different from zero", at the 5% level.

- **If H_0 is not rejected** we say β_j is "statistically insignificant" at the 5% level.

Note the asymmetry in language: we never say we have "accepted" H_0. Failing to reject is not the same as establishing that the parameter is zero.

2.4 A rule of thumb

The procedure above requires consulting a table for every test. For large samples (df > 30) a shortcut is safe, because the critical values barely move:

Degrees of freedom	Two-tailed 5% critical value	One-tailed 5% critical value
30	2.042	1.697
60	2.000	1.671
120	1.980	1.658
∞ (normal)	1.960	1.645

The rules of thumb

Two-sided test at 5%: reject if $|t| > 2$ (or $\beta_j > 2*se$).

One-sided test at 5%: reject if $t > 1.65$ (for $H_1: \beta_j > 0$), or $t < -1.65$ (for $H_1: \beta_j < 0$).

For the one-sided case the sign matters – do not take the absolute value, or you will double the true size of the test.

For small samples use the exact values from the t-table; these approximations are not safe there.

2.5 Testing hypotheses other than zero

If the null specifies some other value a_j:

$$H_0: \beta_j = a_j, \quad t = (\hat{\beta}_j - a_j) / se(\hat{\beta}_j)$$

The general form is worth memorising:

$$t = (\text{estimate} - \text{hypothesised value}) / \text{standard error}$$

Important practical point: the t-statistic reported automatically by EViews tests against zero. For any other null you must compute the statistic yourself, or use a Wald test.

Example: Campus crime and enrollment

Consider a constant elasticity model relating the annual number of crimes on college campuses to student enrollment, so that β_1 is the elasticity of crime with respect to enrollment.

Testing $H_0: \beta_1 = 0$ would be pointless – we obviously expect more crimes on bigger campuses. The interesting hypothesis is that the elasticity equals one, meaning a 1% increase in enrollment leads on average to a 1% increase in crime. The noteworthy alternative is that crime rises more than proportionately:

$$H_0: \beta_1 = 1 \quad \text{versus} \quad H_1: \beta_1 > 1$$

Quantity	Value
Estimated elasticity $\hat{\beta}_1$	1.27
Standard error	0.11
$t = (1.27 - 1)/0.11$	2.46
Critical value at 5%, one-sided, 95 df	1.66
Decision	Reject H_0 in favour of H_1: $\beta_1 > 1$ at the 5% level

The estimate 1.27 points in the direction of the alternative, but pointing in a direction is not enough – the test is what establishes that the evidence is strong enough.

> **An important caveat on this example**
>
> This analysis holds no other factors constant, so the elasticity of 1.27 is not necessarily a good estimate of a ceteris paribus effect. Larger enrollments may be correlated with other causes of crime – bigger schools might sit in higher-crime areas. Controlling for local city crime rates would help.
>
> A statistically significant result from a badly specified model is still a badly specified model.

2.6 The p-value

Choosing a significance level is always somewhat arbitrary – why 5% rather than 1% or 10%? The p-value avoids the arbitrariness by reporting the result of every possible test at once.

The **p-value** is the smallest significance level at which the null hypothesis would be rejected. Equivalently, and more usefully: it is the probability of observing a t-statistic at least as extreme as the one we obtained, if the null hypothesis is true.

Because it is a probability, it always lies between zero and one. Once the p-value is known, a classical test at any level follows immediately:

$$\textit{Reject } H_0 \textit{ if p-value} < \alpha; \quad \textit{otherwise do not reject at the } 100\cdot\alpha\% \textit{ level}$$

For example, a p-value of 0.50 means we would observe a t-statistic as extreme as ours in 50% of all random samples when the null is true – pretty weak evidence against H_0.

2.7 Statistical versus economic significance

These are two different questions, and confusing them is one of the most common errors in applied work.

	Statistical significance	Economic (practical) significance
Determined by	The size of the t-statistic	The size and sign of the coefficient
Question answered	Could this be chance?	Is the effect big enough to matter?
Depends on sample size?	Yes – strongly	No

With large samples, parameters are estimated very precisely: standard errors become small relative to the coefficients, which usually produces statistical significance. Too much focus on significance can then lead to the false conclusion that a variable is "important" for explaining y even though its estimated effect is modest.

One practical response is to tighten the significance level as the sample grows – if you are comfortable with 5% when n is a few hundred, consider 1% when n is a few thousand.

Guidelines for discussing a variable

1. Check for statistical significance. If the variable is significant, discuss the magnitude and sign of the coefficient to convey its practical importance.
2. If it is not significant at the usual levels (10%, 5%, 1%), still ask whether it has the expected effect and whether that effect is practically large. If it is large, compute the p-value. For small samples one can sometimes make a case for p-values as large as 0.20 – though there are no hard rules.
3. A significant variable with an unexpected sign and a practically large effect is troubling and hard to resolve. It usually calls for more thought about the model and the data; a counterintuitive significant estimate often results from the omission of a key variable.

3. Confidence intervals

Confidence intervals – also called interval estimates – provide a range of likely values for the population parameter rather than just a point estimate. Under the CLM assumptions, a 95% confidence interval is:

$$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$$

where c is the 97.5th percentile of the t distribution with n − k − 1 degrees of freedom.

> **What "95% confidence" actually means**
>
> If random samples were obtained over and over again and the interval computed each time, the population value of the parameter would lie inside the interval in 95% of those samples.
>
> The statement is about the procedure across samples, not about the one interval you happen to have computed.

When n − k − 1 > 50, c is so close to 2 that a simple rule of thumb works: $\hat{\beta}_j$ plus or minus two of its standard errors.

3.1 Confidence intervals and two-sided tests are the same thing

Once an interval is constructed, two-tailed tests become trivial. For H_o: $\beta_j = a_j$ against H_1: $\beta_j \neq a_j$, H_o is rejected at the 5% level if and only if a_j lies outside the 95% confidence interval.

In particular, a coefficient is statistically significant at 5% precisely when its 95% interval excludes zero. The interval and the t-test carry identical information – the interval simply displays it in the units of the problem, which usually makes economic significance easier to judge.

> **A confidence interval is only as good as its assumptions**
>
> If important factors correlated with the regressors have been omitted, the coefficient estimates are not reliable: OLS is biased, and the interval is centred in the wrong place.
>
> If heteroskedasticity is present, the standard error is not valid, so the width of the interval is wrong.

> Normality of the errors was also used, though this matters less in applications with hundreds of observations.

4. Testing a single linear combination of the parameters

Sections 2 and 3 dealt with one parameter at a time. Often the hypothesis of interest involves more than one – for instance, whether two coefficients are equal.

Consider comparing the returns to education at junior colleges and four-year colleges (universities), for a population of working people with a high school degree:

$$log(wage) = \beta_0 + \beta_1 jc + \beta_2 univ + \beta_3 exper + u$$

where jc is years attending a two-year college, univ is years at a four-year college, and exper is months in the workforce.

The hypothesis is whether one year at a junior college is worth one year at a university, against the one-sided alternative that it is worth less:

$$H_0: \beta_1 = \beta_2 \quad versus \quad H_1: \beta_1 < \beta_2$$

The trick is to rewrite these in terms of a single quantity, the difference:

$$H_0: \beta_1 - \beta_2 = 0 \quad versus \quad H_1: \beta_1 - \beta_2 < 0$$

Then the usual t-statistic applies to that difference:

$$t = (\hat{\beta}_1 - \hat{\beta}_2) / se(\hat{\beta}_1 - \hat{\beta}_2)$$

The one subtlety is the standard error of the difference, which is not simply the sum of the two standard errors:

$$se(\hat{\beta}_1 - \hat{\beta}_2) = \{ [se(\hat{\beta}_1)]^2 + [se(\hat{\beta}_2)]^2 - 2s_{12} \}^{(1/2)}$$

where s_{12} is an estimate of $Cov(\hat{\beta}_1, \hat{\beta}_2)$. Because the covariance term is needed, this is one case where you cannot do the test from the regression output alone – use a Wald test in EViews. Since the alternative is one-sided, the rejection rule is $t < -c$.

EViews result

Quantity	Value
$\hat{\beta}_1 - \hat{\beta}_2$ (difference in returns)	−0.0102
t-statistic	−1.468
Critical value at 5%, one-sided	−1.645
Two-sided p-value reported by EViews	0.14
One-sided p-value (halve it)	0.07
Decision	Fail to reject H_0: $\beta_1 = \beta_2$

The return to a year at a junior college is estimated to be about one percentage point lower than a year at a university, and the sign is in the direction of the alternative – but the difference is not

statistically significant at 5%. Note the practical trap: EViews reports a two-sided p-value, so for a one-sided test you must halve it.

5. Testing multiple linear restrictions: the F-test

A t-statistic tests one restriction at a time. Frequently we want to test several restrictions jointly – for example, whether a whole group of variables has no effect.

5.1 A joint hypothesis

Consider a model of major league baseball players' salaries:

$$log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

where salary is 1993 total salary, years is years in the league, gamesyr is average games played per year, bavg is career batting average, hrunsyr is home runs per year, and rbisyr is runs batted in per year.

Suppose we want to test that, once years in the league and games per year are controlled for, the three performance statistics have no effect on salary:

$$H_0: \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$$

This is a set of multiple restrictions, and a test of multiple restrictions is called a joint hypotheses test. The appropriate alternative is simply that H_0 is not true – which holds if at least one of β_3, β_4, β_5 differs from zero.

> **Note what the alternative is not**
>
> H_1 is not "all three coefficients are nonzero". It is "at least one is nonzero". Rejecting H_0 therefore does not tell you which variable matters.

5.2 The F-statistic

Estimate two models. The unrestricted model contains everything; the restricted model imposes H_0 by dropping the variables in question:

$$Unrestricted: \ log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + u$$

$$Restricted: \ log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + u$$

The residual sum of squares always increases when variables are dropped, because the restricted model has less freedom to fit. The question is whether the increase is large enough to reject H_0:

$$F = [\ (SSR_r - SSR_{ur}) / q\ ] / [\ SSR_{ur} / (n - k - 1)\ ]$$

where SSR_r and SSR_{ur} are the residual sums of squares from the restricted and unrestricted models, and q is the number of restrictions imposed (here, the number of variables dropped).

Under H_0 and the CLM assumptions:

$$F \sim F_{q,\ n-k-1}$$

Reject H_o in favour of H_1 if $F > c$. If H_o is rejected we say the variables are jointly statistically significant; if not, they are jointly insignificant, which often justifies dropping them.

> **Why is the rejection region one-tailed if the alternative is two-sided?**
>
> F cannot be negative, and it grows only when the restrictions worsen the fit. Any departure from H_o – in any direction – pushes F up. So a single upper tail captures every way the null can fail.

EViews result for the baseball model

Quantity	Value
F-statistic	9.55
Degrees of freedom (q, n − k − 1)	(3, 347)
Critical value at 5% – @qfdist(0.95, 3, 347)	2.63
p-value	0.00
Decision	Reject H_o – the performance statistics are jointly significant

> **The most instructive result in this lecture**
>
> Individually, all three performance variables are statistically insignificant. Jointly, they are overwhelmingly significant. How can both be true?
>
> hrunsyr and rbisyr are highly correlated – good hitters hit home runs and drive in runs. This multicollinearity inflates each variable's standard error, making it hard to identify either partial effect, which is what the individual t-statistics reflect.
>
> The F-test asks a different question: does this group of variables matter as a block? For that question, correlation within the group is much less damaging.
>
> Practical lesson: never drop a group of variables just because none is individually significant. Test them jointly first.

5.3 The F-statistic for overall significance

A special case sets all the slope coefficients to zero – the pessimistic hypothesis that none of the explanatory variables has any effect on y:

$$H_0: \beta_1 = \beta_2 = \ldots = \beta_k = 0$$

Here the restricted model contains only the intercept, and the F-statistic can be written directly in terms of R^2:

$$F = [\, R^2 / k \,] / [\, (1 - R^2) / (n - k - 1) \,]$$

where R^2 is the usual R-squared from the full regression. This is what software reports as the overall F-statistic. If we fail to reject, there is no evidence that any of the independent variables help explain y – which usually means we must look for other variables.

5.4 General linear restrictions

Testing exclusion restrictions is by far the most common use of the F-statistic, but restrictions implied by theory can be more complicated. Consider a house price model:

$$log(price) = \beta_0 + \beta_1 log(assess) + \beta_2 log(lotsize) + \beta_3 log(sqrft) + \beta_4 bdrms + u$$

where assess is the assessed value before sale. Suppose we want to test whether the assessed valuation is rational. Rationality implies two things at once: a 1% change in the assessed value should be associated with a 1% change in price, and the other characteristics should not help explain price once the assessment is controlled for. That is four restrictions:

$$H_0: \beta_1 = 1, \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$$

A Wald test in EViews handles this directly. The resulting p-value is well above conventional levels, so we cannot reject the null – there is essentially no evidence against the hypothesis that assessed values are rational. Note that the first restriction is not an exclusion: $\beta_1 = 1$, not $\beta_1 = 0$. The F-framework accommodates both.

6. Reporting regression results

A few guidelines, useful both for reading published work and for writing your own:

- Report the estimated OLS coefficients, and interpret those on the key variables. Discuss their economic or practical importance, not only their significance.
- Always include the standard errors alongside the coefficients. Some authors prefer t-statistics or p-values; standard errors are marginally preferable because they let the reader construct confidence intervals.
- Always include the R-squared. Reporting the SSR and the standard error of the regression is sometimes useful but not crucial.
- State the number of observations near the estimated equation.
- For one or two models, equation form is fine. For several models, use one or more tables with models in columns.

7. Summary

1. Under the classical linear model assumptions MLR.1 through MLR.6, the OLS estimators are normally distributed.
2. We use t-statistics to test hypotheses about a single parameter, against one- or two-sided alternatives, using one- or two-tailed tests respectively. The most common null is H_0: $\beta_j = 0$, but other values can be tested with the general t-statistic.
3. We choose a significance level, which together with the df and the alternative determines the critical value. It is more informative to compute the p-value – the smallest significance level at which the null would be rejected – so the hypothesis can be assessed at any level.
4. Under the CLM assumptions, confidence intervals can be constructed for each β_j. A two-sided test at the 5% level is equivalent to checking whether the hypothesised value lies outside the 95% interval.
5. For a hypothesis involving two parameters, rewrite it in terms of their difference and apply a t-test – remembering that the standard error of the difference requires the covariance term.

6. The F-statistic tests multiple restrictions. The alternative is two-sided but the rejection region is the upper tail. H_0 is rejected when F exceeds the critical value, or equivalently when the p-value falls below α.
7. Variables can be jointly significant while individually insignificant, and multicollinearity is the usual reason. Always test a group jointly before dropping it.

> **The distinction to carry forward**
>
> A t-test asks: does this one variable matter, holding the others fixed?
>
> An F-test asks: does this group of variables matter, as a block?
>
> Neither asks whether the effect is large enough to care about. That question is economic, and only you can answer it.

Appendix. Asymptotic normality and large-sample inference

If the errors are not normally distributed, the $\hat{\beta}_j$ are not normally distributed either, so t-statistics do not have exact t-distributions and F-statistics do not have exact F-distributions. This is potentially serious, because all our inference relies on critical values and p-values taken from those distributions.

The rescue comes from the central limit theorem. It implies that the OLS estimators satisfy asymptotic normality: they are approximately normally distributed in large enough samples. Consequently t-testing and the construction of confidence intervals proceed exactly as under the CLM assumptions.

This is valuable because MLR.6 has been dropped entirely. The only restriction now placed on the error distribution is that it have finite variance – errors are not "wildly explosive" – which we always assume. Zero conditional mean (MLR.4) and homoskedasticity (MLR.5) are still required.

> **How large is "large enough"?**
>
> There are no general prescriptions. Some econometricians consider n = 30 satisfactory, but this cannot suffice for every possible error distribution. In practice, more than 100 observations is a reasonable working benchmark for cross-sections.

Consistency

Unbiasedness is desirable but cannot always be achieved. Virtually all economists agree that consistency is a minimal requirement: if your estimator of a population parameter is not consistent, you are wasting your time.

An estimator is **consistent** if it converges in probability to the parameter as the sample size grows without bound: plim $\hat{\beta}_j = \beta_j$. Under MLR.1 through MLR.4, OLS is consistent for each parameter – the same four assumptions that deliver unbiasedness.

Consistency involves a thought experiment about what would happen as the sample size gets large, while at the same time obtaining numerous random samples for each sample size. If obtaining more

and more data does not generally get us closer to the parameter value of interest, then we are using a poor estimation procedure.

Graphically, the sampling distribution of $\hat{\beta}_1$ concentrates ever more tightly around the true β_1 as $n_1 < n_2 < n_3$. Note that an estimator can be biased in any finite sample yet still be consistent – the bias must simply vanish as n grows.

> **Granger on consistency**
>
> The Nobel Prize–winning econometrician Clive W. J. Granger once remarked: "If you can't get it right as n goes to infinity, you shouldn't be in this business."

Key terms

Term	Meaning
MLR.6 (normality)	u is independent of the regressors and normally distributed with mean zero and variance σ^2. Implies MLR.4 and MLR.5.
Classical linear model (CLM)	Assumptions MLR.1 through MLR.6 taken together.
Degrees of freedom	n − k − 1: observations minus estimated parameters.
Consistency	plim $\hat{\beta}_j = \beta_j$. The estimator converges to the parameter as n grows. Holds under MLR.1–MLR.4.
t-statistic	(estimate − hypothesised value) / standard error. Measures how many standard errors the estimate lies from the null.
Significance level (α)	The probability of rejecting H_0 when it is true; the size of the test.
Critical value (c)	The cut-off from the relevant distribution, determined by α, the df, and whether the test is one- or two-tailed.
One-tailed test	Rejection region in one tail only; used with a one-sided alternative. The sign of t matters.
Two-tailed test	Rejection region split between both tails; used with H_1: $\beta_j \neq 0$. Based on $\lvert t \rvert$.
p-value	The smallest significance level at which H_0 would be rejected; the probability of a t-statistic at least as extreme as ours if H_0 is true.
Statistical significance	Determined entirely by the size of the t-statistic.
Economic significance	Determined by the size and sign of the coefficient. A separate question from statistical significance.
Confidence interval	$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$. A range of likely values; equivalent to a two-sided test.
Joint hypotheses test	A test of more than one restriction simultaneously.
Restricted / unrestricted model	The model with H_0 imposed / the full model.
F-statistic	Compares the SSR of restricted and unrestricted models, scaled by q and n − k − 1.
Overall significance	The F-test of H_0: all slopes are zero; expressible in terms of R^2.
Wald test	The procedure for testing restrictions, including non-zero and multi-parameter ones.
Asymptotic normality	The OLS estimators are approximately normal in large samples even without MLR.6.

Review questions

Use these to check your understanding before the seminar.

1. Why is MLR.6 needed at all, given that MLR.1–MLR.5 already make OLS BLUE? What specifically does it let us do that Gauss-Markov does not?
2. Why does replacing sd($\hat{\beta}_j$) with se($\hat{\beta}_j$) change the distribution from normal to t? What happens as the sample grows?
3. Give an argument for why wage is unlikely to be conditionally normally distributed. Does this invalidate the t-tests in a sample of 526? Explain, referring to the Appendix.
4. Distinguish unbiasedness from consistency. Can an estimator be biased in every finite sample yet still consistent?
5. In the wage equation, t = 2.39 on exper with 522 df. Carry out the one-sided test at 5% and at 1% (c = 2.326), and state your conclusion in each case.
6. Why is it wrong to use |t| > 1.65 as a rule of thumb for a one-sided test? What is the consequence for the size of the test?
7. The campus crime elasticity is 1.27 with a standard error of 0.11. Test H_o: $\beta_1 = 1$ against H_1: $\beta_1 > 1$. Then explain why the EViews-reported t-statistic cannot be used directly here.
8. A coefficient has p-value 0.08. State whether H_o is rejected at the 10%, 5% and 1% levels, and explain what the 0.08 actually means.
9. A researcher with n = 50 000 reports that a variable is significant at 1% and concludes it is "important". What is the flaw in this reasoning?
10. Show that testing H_o: $\beta_j = 0$ at 5% using a t-statistic and checking whether zero lies in the 95% confidence interval always give the same answer.
11. EViews reports a two-sided p-value of 0.14 for the junior-college test. Explain why 0.07 is the relevant figure, and state the conclusion at 5% and at 10%.
12. In the baseball model, all three performance variables are individually insignificant but jointly significant at any conventional level. Explain how this is possible, and state the practical lesson for model building.
13. Write out the four restrictions implied by "assessed values are rational" in the house price model. Which of them is not an exclusion restriction, and why does that matter for how you run the test?

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 4 and 5 (optional) cover the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapters 3 and 4.